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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 04 – Lecture 02: Supervised Learning (Classification)

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Topic focus: predicting **class labels** and evaluating with proper metrics

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning* (2e), selected chapters.;Scikit-learn documentation, accessed 2026-02-28.

Quick Links

Classification Confusion Metrics Imbalance Summary

Agenda

Classification basics

Confusion matrix

Metrics

Imbalance

Summary

Learning Outcomes

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- Understand the confusion matrix and the meaning of TP/FP/TN/FN.
- Compute and interpret accuracy, precision, recall, and F1.
- Explain thresholds, ROC-AUC intuition, and why imbalance matters.

Abbreviations & Symbols

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- ROC: TPR vs FPR, AUC: area under curve
- Threshold τ: score $\geq \tau \Rightarrow$ predict positive

What is Classification?

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- handwritten digit (0–9)

Common Classification Models (Overview)

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- **SVM**: margin-based classifier (good with clear separation)

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Confusion Matrix (Binary)

	Pred +	Pred -
True +	TP	FN
True -	FP	TN

- TP: correct positive, FP: false alarm
- FN: missed positive, TN: correct negative

Worked Example (Compute Metrics)

Suppose: TP=40, FP=10, FN=20, TN=930 (rare positives).

- Accuracy $= (TP + TN)/(TP + TN + FP + FN)$

High accuracy is easy with many TNs; precision/recall reveal minority-class performance.

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- Recall $= TP/(TP + FN) = 40/60 \approx 0.67$
- F1 $= 2 \cdot \frac{0.80 \cdot 0.67}{0.80 + 0.67} \approx 0.73$

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- AUC summarizes ranking quality (higher is better).

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- Precision-Recall (PR) curve focuses on the positive class.
- For imbalanced problems, PR-AUC is often more informative than ROC-AUC.

Imbalanced Data: Why Accuracy Fails

Example

If 99% emails are not spam, a classifier that always predicts "not spam" gets 99% accuracy.

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If 99% emails are not spam, a classifier that always predicts "not spam" gets 99% accuracy.

- Accuracy can be misleading in imbalance.
- Use precision/recall, F1, and domain-specific costs.
- Lab Experiment-09 focuses on imbalance handling techniques.

Summary

- Classification predicts labels; evaluate with confusion matrix metrics.
- Precision/recall trade-offs depend on error costs.
- Threshold choice matters; ROC-AUC provides a ranking view.
- Imbalance requires careful metrics and techniques.

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Exit Questions

1. Define TP, FP, TN, FN with a medical test example.
2. Why is accuracy misleading on imbalanced datasets?
3. When is recall more important than precision (give one scenario)?
4. What is the role of a classification threshold?